

EXP. NO: 6

DATA VISUALIZATION USING POWER BI

AIM:

To become familiar with the Power BI Desktop workspace and its workflow by importing data from an Excel workbook and a CSV file, cleaning and reshaping it in Power Query, adding calculated columns and DAX measures, and designing an interactive retail dashboard from a variety of visuals.

PROCEDURE:

1. Launch Power BI Desktop and Explore the Workspace

- a. Start Power BI Desktop and close the welcome screen.
- b. Locate the Report, Table, and Model views on the left rail, and the Data, Visualizations, and Filters panes on the right.
- c. Stay in the Report view, which is the canvas used for building the dashboard.

2. Import the Data Sources

- a. Excel data:
 - i. Choose Home → Get Data → Excel Workbook.
 - ii. Browse to RetailSales.xlsx and open it.
 - iii. In the Navigator, tick the Orders sheet.
 - iv. Click Load.
- b. CSV data:
 - i. Choose Home → Get Data → Text/CSV.
 - ii. Browse to SalesTargets.csv and check the preview.
 - iii. Click Load.

3. Clean and Transform the Data in Power Query

- a. Choose Home → Transform Data to open the Power Query Editor.
- b. Select the Orders query and set the OrderDate column to the Date type.
- c. Add a custom column named TotalRevenue with the formula [Quantity] * [UnitPrice].
- d. Add a custom column named YearMonth with the formula Date.ToText([OrderDate], "yyyy-MM").
- e. Set Quantity to Whole Number, and UnitPrice and TotalRevenue to Fixed Decimal Number.
- f. Select the SalesTargets query and set the Month column to the Date type and TargetRevenue to Fixed Decimal Number.
- g. Remove blank rows and duplicate records, then click Close & Apply.

4. Create DAX Measures

- a. The following measures were added to the model:
 - i. Total Revenue = SUM(Orders[TotalRevenue])
 - ii. Units Sold = SUM(Orders[Quantity])
 - iii. Avg Order Value = DIVIDE([Total Revenue], COUNTROWS(Orders))
 - iv. Target Revenue = SUM(SalesTargets[TargetRevenue])

5. Build the Core Charts

- a. Clustered Bar Chart – Revenue by Category:
 - i. Insert a clustered bar chart.
 - ii. Place Category on the Y-axis and Total Revenue on the X-axis.
 - iii. Turn on data labels and sort the bars in descending order of Total Revenue.
- b. Line Chart – Monthly Revenue Trend:
 - i. Insert a line chart.
 - ii. Place YearMonth on the X-axis and Total Revenue on the Y-axis.
 - iii. Name the visual Monthly Revenue Trend.
- c. Donut Chart – Revenue by Region:
 - i. Insert a donut chart.
 - ii. Place Region in Legend and Total Revenue in Values.
 - iii. Name the visual Revenue by Region.

6. Add KPI Cards

- a. Insert three card visuals.
- b. Assign Total Revenue, Units Sold, and Avg Order Value to the cards, one measure each.

7. Build the Top Products Table

- a. Insert a table visual and add Product, Total Revenue, and Units Sold.
- b. In the Filters pane, apply a Top N filter on Product showing the Top 10 items.
- c. Rank the items by Total Revenue.

8. Add the Region Slicer

- a. Insert a slicer visual and add Region to it.
- b. Set the slicer style to Tile so that Central, Coastal, Highland, and Metro appear as buttons.
- c. Click a button to filter every visual on the page for that region.

9. Create the Revenue vs Target Gauge

- a. Insert a gauge visual.
- b. Add Total Revenue to Value and Target Revenue to Target value.
- c. Name the visual Revenue vs Target.

10. Assemble and Save the Dashboard

- a. Place the three KPI cards in a row at the top of the page.
- b. Place the category bar chart and region donut chart beneath the cards.
- c. Place the gauge below the region slicer, on the right-hand side.
- d. Place the monthly trend chart and the top-products table along the bottom.
- e. Align the visuals, set consistent titles and colours, and add a page heading.
- f. Save the report as RetailDashboard.pbix.

OUTPUT:

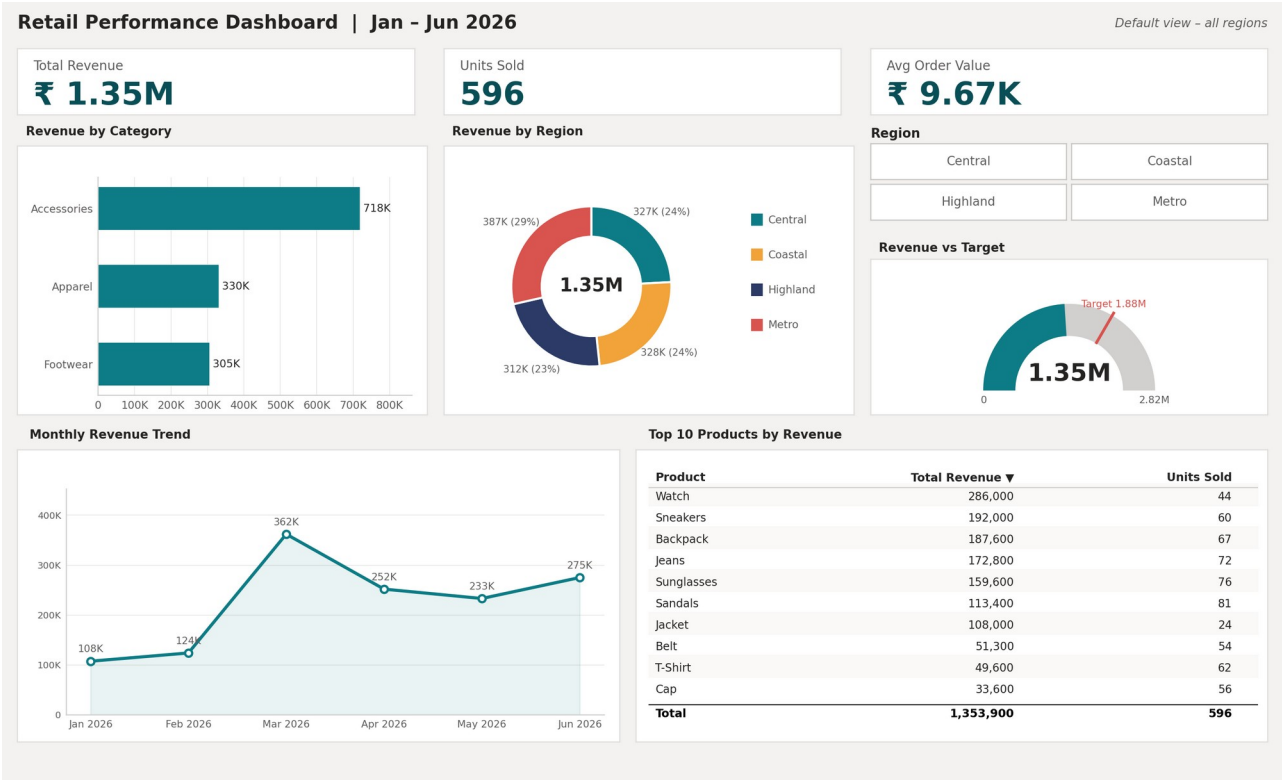


Fig. 1 Complete dashboard with no filter applied (all regions)

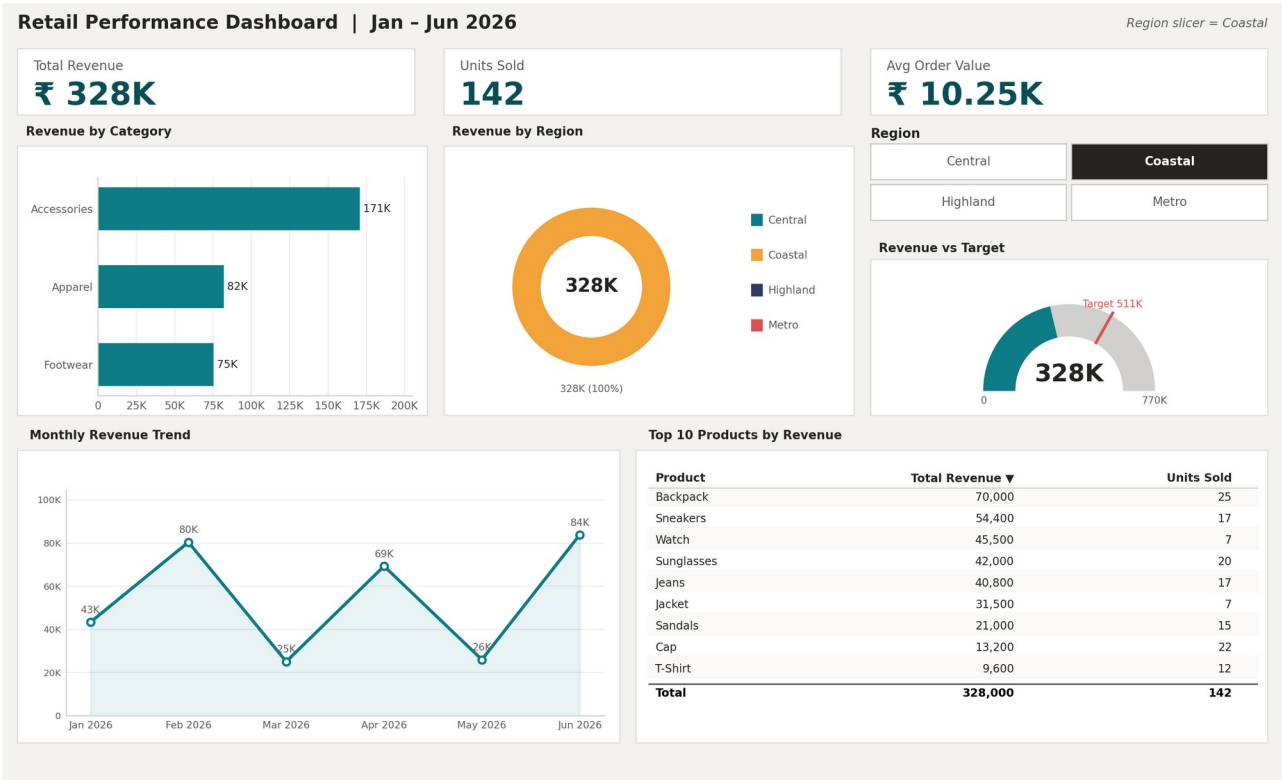


Fig. 2 Dashboard filtered with the Region slicer set to Coastal

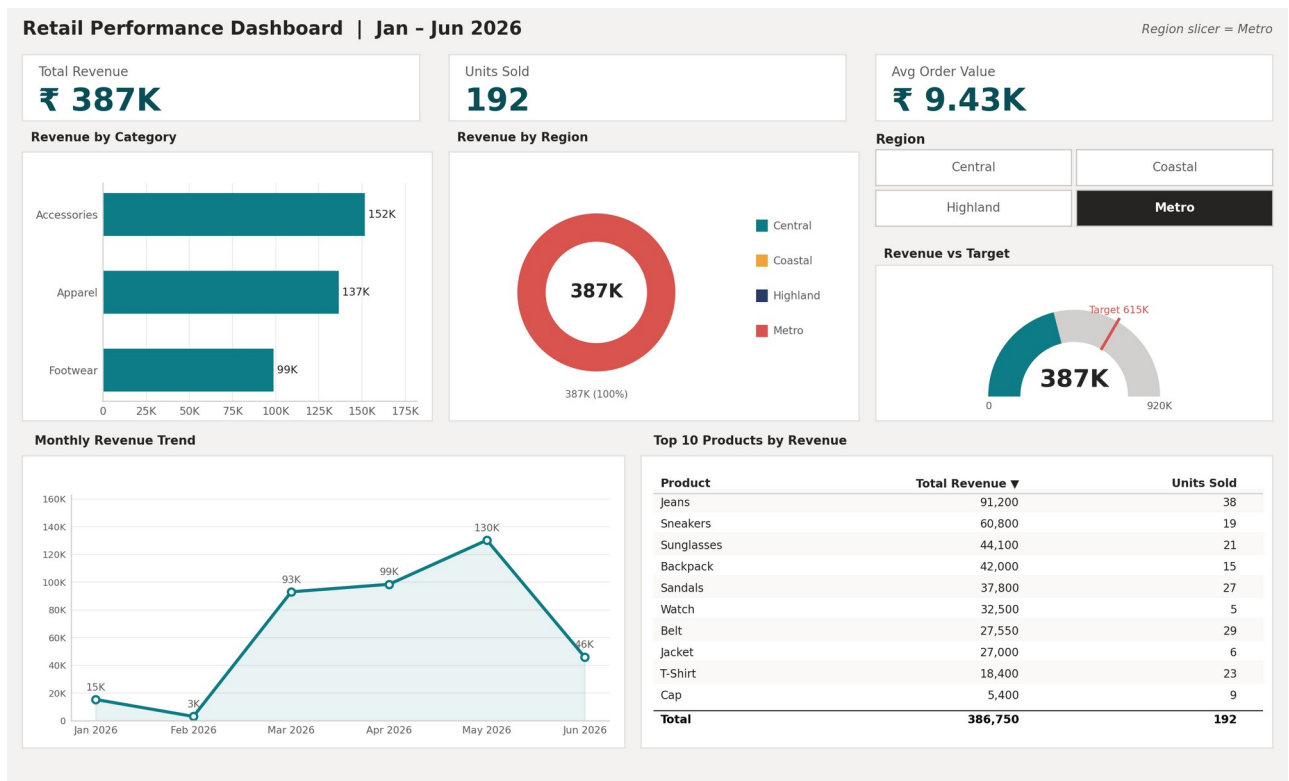


Fig. 3 Dashboard filtered with the Region slicer set to Metro

RESULT:

Thus, the experiment was carried out successfully in Power BI Desktop. Retail sales and target data were imported from Excel and CSV sources, cleaned in Power Query, analysed with DAX measures, and presented as an interactive dashboard with KPI cards, charts, a gauge, a top-products table, and a region slicer.